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High turn down ratio direct control for variable displacement pumps with flow sensing

Abstract

A system includes a variable displacement pump (VDP) in fluid communication with an inlet line and with an outlet line. A bypass valve (BPV) includes a BPV inlet in fluid communication with the outlet line, and a BPV outlet in fluid communication with a bypass line that feeds into the inlet line upstream of the VDP. An actuator is operatively connected to control the BPV to vary flow from the BPV inlet to the bypass line. A flow sensing valve (FSV) is connected in the outlet line. The FSV includes a sensor configured to generate sensor data indicative of flow out of the outlet line. A controller operatively connected to the actuator to control recirculation flow passed through the BPV based on the sensor data and based on a predetermined low threshold of flow through the VDP.

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Background/Summary

BACKGROUND

1. Field

(1) The present disclosure relates to pump control, and more particularly to control for variable displacement pumps (VDPs).

2. Description of Related Art

(2) In a pump, the turn-down ratio is the ratio of the pump's maximum flow to its minimum flow. In fuel delivery systems using a variable displacement pump (VDP), often the pump is subject to a high turn-down ratio. This can drive a pump design with a less than optimal pump efficiency throughout the operating range as a tradeoff for ensuring the turn-down ratio needed. For example, it is beneficial to pump design to minimize this turn-down ratio to be less than 4:1. But large displacement VDPs can struggle to function with such high turn down ratios in typical fuel

metering systems because minimum pump pressure flows to support cooling the VDP are typically higher than the lowest flow for fuel burners, e.g. in gas turbine engines such as on aircraft.

(3) The conventional techniques have been considered satisfactory for their intended purpose. However, there is an ever present need for improved systems and methods for control of VDPs. This disclosure provides a solution for this need.

SUMMARY

(4) A system includes a variable displacement pump (VDP) in fluid communication with an inlet line and with an outlet line. The VDP includes a variable displacement mechanism configured to vary pressure to the outlet line. A bypass valve (BPV) includes a BPV inlet in fluid communication with the outlet line, and a BPV outlet in fluid communication with a bypass line that feeds into the inlet line upstream of the VDP. An actuator is operatively connected to control the BPV to vary flow from the BPV inlet to the bypass line. A flow sensing valve (FSV) is connected in the outlet line. The FSV includes a sensor configured to generate sensor data indicative of flow out of the outlet line. A controller operatively connected to the actuator to control recirculation flow passed through the BPV based on the sensor data and based on a predetermined low threshold of flow through the VDP. The BPV outlet can be the only outlet of the BPV so all flow through the BPV is supplied to the BPV outlet.

(5) The controller can be configured to control the BPV to maintain a baseline flow through the BPV under a first condition wherein requested flow from the downstream system is above than the predetermined low threshold. The controller can be configured to control the BPV to increase the flow through the BPV above the baseline flow for a second flow condition wherein requested flow from the downstream system is at or below the predetermined low threshold.

(6) A first electrohydraulic servo valve (EHSV) can be connected in fluid communication with the BPV by a first control line. The first EHSV can be connected in fluid communication with both the inlet line and with the outlet line through respective connection lines. The first EHSV can be operatively connected to the controller for active control of the first EHSV to actuate the BPV. A first position sensor can be operatively connected to the BPV to provide sensor output indicative of position of a valve member of the BPV. The first position sensor can be operatively connect the controller to provide feedback for controlling the BPV. A second EHSV can be connected in fluid communication with the variable displacement mechanism by a second control line for control of flow through the VDP. The second EHSV can be connected in fluid communication with both the inlet line and with the outlet line through respective connection lines. The second EHSV can be operatively connected to the controller for active control of the second EHSV to actuate the variable displacement mechanism. A second position sensor can be operatively connected to the variable displacement mechanism to provide sensor output indicative of position of the variable displacement mechanism. The second position sensor can be operatively connect the controller to provide feedback for controlling the variable displacement mechanism.

(7) A pressure sensor can be operatively connected to the outlet line to generate sensor output indicative of pressure in the outlet line. The pressure sensor can be operatively connected to the controller for active control of the variable displacement mechanism and/or of the BPV based on pressure in the outlet line. The sensor output can be indicative of flow demanded by the downstream system supplied by the outlet line. The controller can be configured to control position of the valve member of the BPV to maintain bypass flow through the BPV in the second condition wherein the controller governs the bypass flow through the BPV according to

(8) $BF = PF - DSFD$

wherein BF is flow through the BPV, PF is flow through the VDP, and DSFD is flow demanded by the downstream system supplied by the outlet line as indicated by the sensor output.

(9) The FSV can include an FSV inlet, an FSV outlet, and a valve member. A biasing member can bias the valve member in a first direction. Pressure of flow through the FSV from the FSV inlet to the FSV outlet can bias the valve member in a second direction opposite the first direction. The

sensor can include a position sensor operatively connected to monitor position of the valve member in the FSV to generate the sensor data. The FSV can include a pressure port on a side of the valve member opposite from the FSV inlet and the FSV outlet. A pressure line can connect the FSV outlet in fluid communication with the pressure port.

(10) A method includes receiving sensor feedback from a flow sensing valve (FSV) indicative of flow demanded by a downstream system supplied from an outlet line of a variable displacement pump (VDP). The method includes controlling a bypass valve (BPV) to recirculate flow from the outlet line to an input line of the VDP in the event of the sensor feedback dropping below a predetermined low threshold of flow through the VDP.

(11) The method can include controlling the BPV to recirculate flow from the outlet line to the inlet line at a constant base recirculation rate in the event of flow demanded by the downstream system being at or above the predetermined low threshold of flow through the VDP. The base recirculation rate can be zero recirculation flow. The method can include receiving pressure data from a pressure sensor in the outlet line. Controlling the BPV to recirculate flow can include controlling the BPV based at least in part on the pressure data.

(12) Controlling the BPV to recirculate flow can include governing the bypass flow through the BPV according to

(13) $BF = PF - DSFD$

wherein BF is flow through the BPF, PF is flow through the VDP, and DSFD is flow demanded by the downstream system supplied by the outlet line as indicated by the sensor feedback. The method can include receiving data from a sensor indicative of position of a valve member of the BPV, wherein controlling the BPV includes controlling the BPV based on position of the valve member.

(14) These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1 is a schematic view of an embodiment of a system constructed in accordance with the present disclosure, showing the connections of the variable displacement pump (VDP) and a bypass valve (BPV).

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(3) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an embodiment of a system in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. The systems and methods described herein can be used to provide high turn down ratios for variable displacement pumps, and for control of the same, such as for use in supplying fuel to gas generators in aircraft engines.

(4) The system **100** includes a variable displacement pump (VDP) **102** in fluid communication with an inlet line **104** and with an outlet line **106**. The VDP **102** includes a variable displacement mechanism **108** configured to vary pressure to the outlet line **106**. A bypass valve (BPV) **110** includes a BPV inlet **112** in fluid communication with the outlet line **106**, and a BPV outlet **114** in fluid communication with a bypass line **116** that feeds into the inlet line **104** upstream of the VDP **102**. An actuator **118** is operatively connected to control the BPV **110** to vary flow from the BPV

inlet **112** to the bypass line **116**. A flow sensing valve (FSV) **148** is connected in the outlet line **106**. The FSV **148** includes a sensor **150**, such as a linear variable differential transformer (LVDT), configured to generate sensor data indicative of flow out of the outlet line **106** to the downstream system **122** supplied by the outlet line **106**.

(5) A controller **120** is operatively connected to the actuator **118** to control recirculation flow passed through the BPV **110** based on the sensor data and based on a predetermined low threshold of flow through the VDP **102**. The downstream system **122** can be a combustor, augmentor, or other gas generator of a gas turbine engine, for example. The low threshold of flow of the VDP **104** can be the threshold below which the VDP **104** cannot self-lubricate, or other design requirements for low or minimum flow. The BPV outlet **114** can be the only outlet of the BPV **110** so all flow through the BPV **110** from the BPV inlet **112** is supplied to the BPV outlet **114**.

(6) The controller **120** is configured, e.g. including analog circuitry, digital logic, and/or machine readable instructions, to control the BPV **110** to maintain a baseline flow through the BPV **110** under a first condition wherein requested flow from the downstream system **122** is above the predetermined low threshold. The controller **120** is configured to control the BPV **110** to increase the flow through the BPV **110** above the baseline flow for a second flow condition wherein requested flow from the downstream system **122** is at or below the predetermined low threshold.

(7) The actuator **118** includes a first electrohydraulic servo valve (EHSV) **124** that is connected in fluid communication with the BPV **110** by a first control line **126**. The first EHSV **124** is connected in fluid communication with both the inlet line **104**, by way of the recirculation line **116**, and with the outlet line **106** through respective connection lines **128**, **130**. The first EHSV **124** is operatively connected to the controller **120** for active control of the first EHSV **124** to actuate the BPV **110**.

(8) A first position sensor **132**, such as a linear variable differential transformer (LVDT), is operatively connected to the BPV **110** to provide sensor output indicative of position of a valve member **134** of the BPV **110** within the BPV **110**. The first position sensor **132** is operatively connect the controller **120** to provide feedback for controlling the BPV **110**. A second EHSV **136** is connected in fluid communication with the variable displacement mechanism **108** by a second control line **138** for control of flow through the VDP **102**. The second EHSV **136** is connected in fluid communication with both the inlet line **104** and with the outlet line **106** through respective connection lines **140**, **142**. The second EHSV **136** is operatively connected to the controller **120** for active control of the second EHSV **136** to actuate the variable displacement mechanism **108** to control flow through the variable displacement mechanism **108**. A second position sensor **144**, such as an LVDT, is operatively connected to the variable displacement mechanism **108** to provide sensor output indicative of position of the variable displacement mechanism **108**, wherein the second position sensor **144** is operatively connect the controller **120** to provide feedback for controlling the variable displacement mechanism **108**.

(9) A pressure sensor **146** is operatively connected to the outlet line **106** to generate sensor output indicative of pressure in the outlet line **106**. The pressure sensor **146** is operatively connected to the controller **120** for active control of the variable displacement mechanism **108** and/or of the BPV **110** based on pressure in the outlet line **106**. The sensor output of sensor **150** is indicative of flow demanded by the downstream system **122**. The controller **120** is configured to control position of the valve member **134** of the BPV **110** to maintain bypass flow through the BPV in the second condition, i.e. when recirculation through the BPV **110** is needed because flow demanded by the downstream system **122** drops below the predetermined low threshold for flow through the VDP **102**. In this second condition, the controller **120** governs the bypass flow through the BPV **110** according to

(10) $BF = PF - DSFD$

wherein BF is flow through the BPV **110**, PF is flow through the VDP **102** (e.g. as indicated by sensors **144** and/or **146**), and DSFD is flow demanded by the downstream system **122** as indicated by the sensor output of the sensor **150**. In the first condition, e.g. when flow demanded by the

downstream system **122** is at or above the predetermined low threshold for flow through the VDP **112**, the controller **120** controls the BPV **110** to recirculate flow from the outlet line **106** to the inlet line **102** at a constant base recirculation rate. The base recirculation rate can be zero recirculation flow.

(11) The FSV **148** includes an FSV inlet **152**, an FSV outlet **154**, and a valve member **156** that moves inside the FSV **148** based on how much pressure is acting on the valve member **156**. A biasing member **158** biases the valve member **156** in a first direction, e.g. the biasing member **160** can bias the valve member to the left as oriented in FIG. **1**. Pressure of flow through the FSV **148** from the FSV inlet **152** to the FSV outlet **154** biases the valve member **156** in a second direction opposite the first direction, e.g. to the right as oriented in FIG. **1**. The sensor **150** includes a position sensor operatively connected to monitor position of the valve member **156** in the FSV **148** to generate the sensor data indicative of that position. The FSV **148** includes a pressure port **162** on a side of the valve member **156** opposite from the FSV inlet **152** and the FSV outlet **154**. A pressure line **164** connects the FSV outlet **154** in fluid communication with the pressure port **162**. The pressure port **162** of the FSV **148** connects the backside of the FSV **148** to the downstream pressure in the outlet line **106**, via the pressure line **164**. The pressure balance on the FSV **148** is pump outlet pressure versus the sum of pressure from FSV outlet pressure (via pressure line **164**) and from the spring force from biasing member **160**. The biasing member **160** creates a constant pressure drop across the FSV **148** to allow for the controller **120** to calculate flow across the FSV **120**.

(12) Systems and methods as disclosed herein provide various potential benefits including the following. There can be a reduction in valve count and the need for a metering valve can be eliminated. The metering system can be faster than legacy systems, e.g. where a metering valve controls pressure control valve (PCV) which controls pump displacement or the like.

(13) The methods and systems of the present disclosure, as described above and shown in the drawings, provide for high turn down ratios for variable displacement pumps, and for control of the same, such as for use in supplying fuel to gas generators in aircraft engines. While the apparatus and methods of the subject disclosure have been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A method comprising: Receiving sensor feedback from a flow sensing valve (FSV) indicative of flow demanded by a downstream system supplied from an outlet line of a variable displacement pump (VDP); operating an electrohydraulic servo valve (EHSV) such that a pressure of a control line for a bypass valve (BPV) is lower in response to the sensor feedback dropping below the predetermined low threshold of flow through the VDP; opening the BPV to redirect flow from the pump outlet line to a system inlet line in reaction to the pressure of the control line being lower; and controlling a second EHSV such that a second pressure of a second control line for a variable displacement mechanism of the VDP is varied, and thereby varying a pressure of the pump outlet line.
2. The method as recited in claim 1, wherein the flow from the outlet line to the inlet line is recirculated at a constant base recirculation rate in response to the flow demanded by the downstream system being at or above the predetermined low threshold of flow through the VDP.
3. The method as recited in claim 2, wherein the base recirculation rate is zero recirculation flow.
4. A system comprising: a variable displacement pump (VDP) fluidically connected to an inlet line and fluidically connected to a pump outlet line, wherein the VDP comprises a variable displacement mechanism configured to vary pressure to the pump outlet line; a bypass valve (BPV) comprising a BPV inlet fluidically connected with the pump outlet line, and a BPV outlet

fluidically connected to a bypass line that feeds into the inlet line; an actuator comprising a first electrohydraulic servo valve (EHSV), wherein the first EHSV is fluidically connected to the BPV by a first control line; a lower-pressure line fluidically connecting the first EHSV to the inlet line; a higher-pressure line fluidically connecting the first EHSV to the pump outlet line; a flow sensing valve (FSV) fluidically connected to the pump outlet line, wherein the FSV comprises a sensor configured to generate sensed flow out of the pump outlet line; a controller connected to the first EHSV and configured to control recirculation flow through the BPV based on the sensed flow out of the pump outlet line and based on a predetermined low threshold of flow through the VDP, wherein: the controller is configured to: control the BPV to maintain a baseline flow through the BPV during a first flow condition in which requested flow from the downstream system is above the predetermined low threshold; and control the BPV to increase flow through the BPV above the baseline flow during a second flow condition in which the requested flow from the downstream system is at or below the predetermined low threshold; and a first position sensor connected to the BPV to provide sensor output indicative of a position of a valve member of the BPV, wherein the first position sensor is connected to the controller to provide a first feedback for controlling the BPV; a second electrohydraulic servo valve (EHSV) fluidically connected to the system inlet, fluidically connected to the pump output line, and fluidically connected with the variable displacement mechanism via a second control line, and wherein the controller is configured to actuate the variable displacement mechanism via the second EHSV.

5. The system of claim 4, wherein: a second position sensor is connected to the variable displacement mechanism; the second position sensor is connected to the controller; the second position sensor is configured to provide a sensed variable displacement mechanism position to the controller; and the controller is configured to respond to the sensed variable displacement mechanism position by controlling the variable displacement mechanism.

6. The system of claim 5, and further comprising a pressure sensor, wherein: the pressure sensor is connected to the pump outlet line; the pressure sensor is connected to the controller; the pressure sensor is configured to provide a sensed pressure of the pump outlet line to the controller; the controller is configured to respond to the sensed pressure of the pump outlet line by controlling the variable displacement mechanism and/or the BPV.

7. A system comprising: a system inlet; a system outlet downstream of the system inlet; a pump inlet line fluidically connected to the system inlet; a variable displacement pump (VDP) comprising: a variable displacement mechanism; a pump inlet fluidically connected to the pump inlet line; and a pump outlet fluidically connected to a pump outlet line; a bypass valve (BPV) comprising: a BPV inlet fluidically connected to the pump outlet line downstream of the pump outlet; a BPV outlet downstream of the BPV inlet; and a valve member configured to control flow between the BPV inlet and the BPV outlet; a bypass line fluidically connecting the BPV outlet to the pump inlet line; a flow sensing valve (FSV) fluidically connected to the pump outlet line and downstream of the pump outlet line, wherein the FSV comprises a FSV sensor configured to provide a sensed output flow of the system outlet line to a controller; an actuator comprising a first electrohydraulic servo valve (EHSV), wherein the first EHSV comprises a first port, a second port, and a third port; a lower-pressure line fluidically connected to the system inlet line and fluidically connected to the first port of the first EHSV; a higher-pressure line fluidically connected to the pump outlet line and fluidically connected to the second port of the first EHSV; and a first control line fluidically connecting the third port of the first EHSV to a control chamber of the BPV; a first position sensor connected to the BPV and connected to the controller, wherein the first position sensor is configured to provide a sensed valve member position to the controller; and a second EHSV connected to the controller, wherein the second EHSV comprises a fourth port, a fifth port, and a sixth port; wherein: the fourth port is fluidically connected to the system inlet; the fifth port is fluidically connected to the pump output line; the sixth port is fluidically connected with the variable displacement mechanism via a second control line; and the controller is configured to

actuate the variable displacement mechanism via the second EHSV; wherein the first EHSV is configured to vary flow from the BPV inlet to the BPV outlet by controlling a pressure of the first control line; wherein the controller is configured to actuate the variable displacement mechanism via the second EHSV; and wherein the controller is configured to open the first port of the first EHSV and close the second port of the first EHSV under a first flow condition wherein a requested flow from a downstream system is above a predetermined low threshold.

8. The system as recited in claim 7, wherein the controller is configured to close the first port of the EHSV and open the second port of the EHSV under a second flow condition wherein a requested flow from a downstream system is at or below the predetermined low threshold.

9. The system as recited in claim 7 and further comprising a second position sensor connected to the controller, wherein the second position sensor is connected to the variable displacement mechanism and wherein the second position sensor is configured to provide a variable displacement mechanism position to the controller.

10. The system as recited in claim 9, and further comprising a pressure sensor connected to the controller, wherein the pressure sensor is connected to the pump outlet line and wherein the pressure sensor is configured to provide a sensed pump outlet line pressure.

11. The system as recited in claim 10, wherein: the sensed output flow of the system outlet line is an indicator of the requested flow; the controller is configured to control a position of the valve member of the BPV to maintain bypass flow through the BPV under the second flow condition; and the controller governs the bypass flow through the BPV according to $BF = PF - DSFD$, wherein BF is the bypass flow through the BPV, PF is flow through the VDP, and DSFD is the flow demanded by the downstream system.

12. The system as recited in claim 7, wherein the FSV comprises: an FSV inlet; an FSV outlet; a biasing member; and a FSV valve member, wherein the biasing member biases the FSV valve member in a first direction.

13. The system as recited in claim 12, wherein the FSV comprises a pressure port on a side of the FSV opposite from the FSV inlet and the FSV outlet, wherein the FSV valve member is between the FSV inlet and the pressure port, and wherein a pressure line fluidically connects the FSV outlet to the pressure port.
